

Homework 6 (Extra - not graded)

MATH 518 - Fall 2025

Unless stated otherwise, k is an algebraically closed field of characteristic zero.

1. What is $\dim(X)$ in the following cases?

1. $X = V(xy - 1) \subset k^2$,
2. $X = V(xy - z^2, x + w - 1) \subset k^4$
3. $X = \mathbb{P}^n(k)$
4. $X = V(xy - w^2, xz - w^2) \subset \mathbb{P}^2(k)$.

2. (a) Let $K \subseteq K'$ be two finitely generated field extension of k . Show that $\text{tr.deg}_k(K) \leq \text{tr.deg}_k(K')$.

(b) Let $f: X \dashrightarrow Y$ be a dominant rational map between quasi-projective. Show that $\dim(Y) \leq \dim(X)$. Give an example where $\dim(Y) < \dim(X)$.

If it is a finite morphism, then $f(X)$ is dominant by definition. In that case, the condition that $k[X]$ is integral over $k[Y]$ implies that $k(X)$ is an algebraic extension of $k(Y)$, so $\dim(X) = \dim(Y)$.

3. (a) Let $f, g: X \rightarrow Y$ be morphisms between two quasi-projective varieties X and Y . If f and g agree on U , show that they agree on X .

(b) Verify that the relation that we used to define a rational map is an equivalence relation.

4. Let $\phi: \mathbb{P}^2(k) \dashrightarrow \mathbb{P}^3(k)$ be the rational map represented by

$$[x : y : z] \longmapsto [xy : xz : y^2 : yz].$$

Determine the domain U where ϕ is defined. Show that $\phi: \mathbb{P}^2(k) \dashrightarrow X$ is birational, where $X = V(z_1z_4 - z_2z_3)$. Deduce that X is the closure of $f(U)$.

5. Let $X \subseteq k^2$ be the curve $X = V(f)$ where $y^2 - x^3(x - 1)^2$ in k^2 .

- (a) Determine the points where X is smooth.
- (b) Use a sequence of blow-ups to find a smooth curve Y in k^2 that is birational to Y . Write the birational map and determine its inverse.
- (c) Deduce that X is a rational curve, and find a birational map from $X \dashrightarrow k$.